

PRAVEEN EMANI

SENIOR SOFTWARE ENGINEER · AGENTIC AI SYSTEMS · MLOPS

Senior Software Engineer with 7+ years building production agentic AI and ML systems end-to-end — autonomous multi-agent workflows (LangGraph), reliable RAG and tool-calling systems, LLM-as-judge evaluation, and safety guardrails. Strong foundation in classical ML, distributed systems (M.S.), and full-stack engineering.

WORK EXPERIENCE

AI Crew Solutions

Aug 2022 – Present

Senior Software Engineer — Agentic AI & Platform Team · Wichita, KS (Hybrid)

- Architected and led 5+ production autonomous multi-agent systems (LangGraph, AutoGen) with supervisor-router patterns, hierarchical delegation, and ReAct + reflection; achieved 85%+ autonomous resolution on enterprise workflows.
- Designed end-to-end RAG pipelines (hybrid retrieval, reranking, query planning, citation grounding) with LlamaIndex + Pinecone / pgvector; cut hallucination ~60% and lifted faithfulness to 92%+.
- Built production tool / function-calling infrastructure (Pydantic schemas, multi-tool chaining, sandboxing, fallbacks) across 25+ APIs; <0.1% unsafe invocation via guardrails and human-in-the-loop gates.
- Productionized agent evaluation frameworks (golden datasets, multi-axis rubrics, LLM-as-judge with human calibration, CI-gated regression) on Braintrust / LangSmith; reduced production incidents 45%.
- Owned the MLOps / AgentOps platform (Docker / K8s + KubeRay, GitOps CI/CD, MLflow, LiteLLM / vLLM multi-model routing); delivered 3–5x throughput and 40–60% cost reduction at high volume.
- Led a team of 6: defined architecture standards and internal SDKs, ran design / code reviews, mentored on prompt engineering and reliability; +35% team velocity, >85% test coverage.

AI Crew Solutions

Jun 2019 – Aug 2022

Software Engineer — Full-Stack & ML Systems · Wichita, KS

- Built full-stack features (Node.js / Express, Angular, MongoDB, Python microservices) and shipped classical ML models (scikit-learn, TensorFlow, XGBoost, Hugging Face) to production with 20–30% metric lifts.
- Designed ETL / feature pipelines in Python, set up CI/CD, Docker, and basic Kubernetes on AWS; mentored juniors on backend and ML integration.

PERSONAL PROJECTS

Telegram Email Automation Agent — LangGraph + HITL

- LangGraph supervisor orchestration (conditional routing, hierarchical delegation) for summarizing, drafting, prioritizing, and acting on email, with Telegram Human-in-the-Loop approvals for safe autonomy.
- Hybrid persistent memory, guardrailed Pydantic tools, LangSmith tracing, and LLM-as-judge evals on golden datasets; FastAPI + Docker, open-sourced as a reusable template.

LangGraph + FastAPI Agent Backend Template

- Async LangGraph execution with persistent checkpoints, tool registry, secure endpoints, observability (LangSmith + OpenTelemetry), guardrails, rate limiting, and sandboxed tools.
- Docker + CI/CD with session management and error recovery; powers the Telegram Email Agent.

LangGraph Agent Evaluation & Trajectory Harness

- Golden-dataset + LLM-as-judge pipeline with trajectory analysis (task success, reasoning, tool usage, safety), offline/online evals, and CI/CD regression with trace visualization.
- Applied across multiple agents to improve reliability; open-sourced.

Reusable LangGraph Multi-Agent Patterns Library

- Supervisor, hierarchical delegation, reflection, HITL handoff, and memory modules with Pydantic state schemas, plus documented graphs, Mermaid diagrams, and guardrails.
- Reusable observability hooks for common patterns (ReAct, plan-and-execute); generalized from production agents and maintained as open source.

CONTACT

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SKILLS

Languages & Frameworks

Python · TypeScript · JavaScript · SQL · Node.js · React · Angular

Agentic AI & LLM

LangGraph · LlamaIndex · AutoGen · CrewAI · RAG · Multi-agent orchestration · Tool calling · LLM-as-judge · Guardrails

ML & Modeling

PyTorch · TensorFlow · scikit-learn · Hugging Face · XGBoost · Embeddings · Vector search · LoRA · QLoRA

Infrastructure & MLOps

AWS · SageMaker · Bedrock · Docker · Kubernetes · GitHub Actions · ArgoCD · MLflow · LangSmith · Grafana

EDUCATION

M.S. — Computer Science and Engineering

Wichita State University, USA

B.Tech — Electronics & Computer Engineering

RGUKT, RK Valley, India

CERTIFICATIONS

NVIDIA

Certified Professional — Agentic AI (NCP-AAI)

DeepLearning.AI

Generative AI with Large Language Models

DeepLearning.AI

Deep Learning Specialization

WRITING & BLOGGING

Designing Production-Grade Agentic Systems with LangGraph

Beyond Vector Search: Evaluating RAG Pipelines

Safe Tool Calling: Guardrails & Human-in-the-Loop for LLM Agents